



**Department of Electrical and Computer Engineering
North South University**

Directed Research

Sycophancy under social pressure in LLMs

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LETTER OF TRANSMITTAL

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To

Dr. Rajesh Palit

Chairman,

Department of Electrical and Computer Engineering

North South University, Dhaka

Subject: Submission of Capstone Project Report on “Sycophancy under social pressure in LLMs”

Dear Sir,

With due respect, I would like to submit my Research Project Report on “Sycophancy under social pressure in LLMs” as a part of my BSc program. This project was very much valuable to us as it helped us gain experience from practical field and apply in real life.

I will be highly obliged if you kindly receive this report and provide your valuable judgment. It would be our immense pleasure if you find this report useful and informative to have an apparent perspective on the issue.

Sincerely Yours,

Afia Mubassira Ali Raisa

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DECLARATION

This is to declare that this project is my original work. No part of this work has been submitted elsewhere partially or fully for the award of any other degree or diploma. All project related information will remain confidential and shall not be disclosed without the formal consent of the project supervisor. Relevant previous works presented in this report have been properly acknowledged and cited. The plagiarism policy, as stated by the supervisor, has been maintained.

ACKNOWLEDGEMENTS

The author would like to express their heartfelt gratitude towards their project and research supervisor, Dr. Mohammad Shifat-E-Rabbi , Assistant Professor, Department of Electrical and Computer Engineering, North South University, Bangladesh, for his invaluable support, precise guidance and advice pertaining to the experiments, research and theoretical studies carried out during the course of the current project and also in the preparation of the current report.

Furthermore, the author would like to thank the Department of Electrical and Computer Engineering, North South University, Bangladesh for facilitating the research.

ABSTRACT

Large Language Models (LLMs) are increasingly being used for education, research, decision support, and everyday information retrieval. However, their tendency to agree with users even when the user's claims are incorrect presents a significant reliability concern. This behavior, known as sycophancy, occurs when a model prioritizes agreement with the user's stated position over independent and factually correct reasoning. This research investigates the susceptibility of multiple LLMs to sycophancy under different forms of social pressure and evaluates a lightweight prompting-based mitigation strategy.

A controlled dataset of 270 factual questions across 18 knowledge categories was developed, with each question presented under six user-tone conditions: Neutral, Mild disagreement, Strong disagreement, Angry/Frustrated pushback, Flattering pressure, and Confident/Assertive correction. The same underlying factual questions and incorrect user claims were used across the tone conditions to isolate the effect of social pressure. The dataset was evaluated across nine model configurations from three model families: GPT-5.6 Sol, Claude Opus 5, and three Gemini variants, with multiple reasoning or effort levels tested where available. In total, 16,200 model responses were collected and evaluated for both correctness and sycophancy.

The results demonstrate substantial variation in sycophancy susceptibility across models. Claude Opus 5 showed the lowest overall sycophancy rates, while Gemini 3.1 Pro exhibited the highest rate among the evaluated configurations. GPT-5.6 Sol showed a clear reduction in sycophancy as reasoning effort increased, whereas Claude Opus 5 showed little meaningful change across effort levels. User tone also influenced model behavior, with confident and mild disagreement conditions frequently acting as strong social-pressure cues.

Finally, an inference-time anti-sycophancy system instruction was tested on all three Gemini variants. The mitigation reduced sycophancy from 17.35% to 6.98% for Gemini 3.1 Pro, from 16.91% to 8.02% for Gemini 3.6 Flash, and from 6.05% to 5.00% for Gemini 3.5 Flash-Lite, without reducing accuracy. These findings demonstrate that sycophancy varies considerably across models and interaction conditions, while a simple user-side prompting intervention can substantially reduce the behavior without requiring model retraining or fine-tuning.

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Chapter 1

Introduction

1.1 Background and Motivation

Large language models (LLMs) are now used as tutors, research assistants, coding partners, and general-purpose decision-support tools, often in settings where the user cannot easily verify the model's answer. This creates a specific reliability risk that is distinct from a model simply being wrong: a model can know the correct answer and still abandon it, or reinforce a user's incorrect claim purely because of how the claim is presented — confidently, angrily, flatteringly, or repeatedly. This behaviour is generally referred to as sycophancy: the tendency of a model to prioritize agreement with the user, or the user's apparent preferences, over independent, truthful, well-calibrated judgement.

Sycophancy has been documented as a measurable phenomenon in prior alignment research. Sharma et al. (2023), in "Towards Understanding Sycophancy in Language Models," showed that models trained with human feedback can learn to match a user's stated beliefs even when those beliefs are incorrect, because agreement is often rated more highly by human evaluators than a correction would be. Follow-up studies such as SycEval (2025) and the ELEPHANT study on social sycophancy (2025) have since reported sycophancy rates in the range of roughly half of all evaluated responses for some general-purpose models under adversarial user framing, and have shown that the effect appears across multiple model providers rather than being specific to one vendor. These findings motivate a more systematic question: does the rate of sycophancy vary in a predictable way with the model family used, the amount of reasoning effort the model is allowed to apply, and the specific social tone the user adopts — and, if so, can a practical, low-cost intervention reduce it.

The motivation for this project is therefore twofold. First, as LLMs are increasingly deployed in domains such as education, health information, and everyday fact-checking, an assistant that concedes a correct answer under social pressure can actively spread or reinforce misinformation rather than correct it, which is a more insidious failure mode than an assistant that is simply wrong on its own. Second, most publicly available sycophancy research evaluates one model or one provider at a time; there is comparatively little controlled, apples-to-apples data comparing sycophancy rates across model families, reasoning/effort settings, and user-tone conditions using the exact same question set. This project is designed to close part of that gap and to test a concrete, deployable mitigation rather than only measuring the problem.

1.2 Purpose and Goal of the Project

The purpose of this project is to build a controlled, repeatable experimental framework for measuring LLM sycophancy and to evaluate a lightweight mitigation strategy against it. Concretely, the project has four goals:

First, to construct a shared, factual question bank spanning a broad range of knowledge categories, each with six user-tone variants (Neutral, Mild disagreement, Strong disagreement, Angry/Frustrated pushback, Flattering pressure, and Confident/Assertive

correction) so that tone can be isolated as an experimental variable while the underlying factual content of each question stays fixed.

Second, to run this identical question bank against multiple current-generation model families OpenAI's GPT-5.6 Sol, Anthropic's Claude Opus 5, and three variants of Google's Gemini model family at multiple reasoning/effort levels where available, producing a directly comparable, cross-provider sycophancy dataset.

Third, to quantify sycophancy rates along several dimensions overall, by reasoning/effort level, by user tone, and by knowledge category in order to identify which conditions and topics most reliably induce sycophantic behaviour and whether that susceptibility tracks model scale, model family, or reasoning effort.

Fourth, to design and test a mitigation: a system-level instruction that asks the model to reason independently of the user's stated position and to resist a named list of social-pressure cues before answering, and to measure whether this intervention meaningfully reduces sycophancy without degrading answer accuracy. The project's explicit contribution is not merely to document that sycophancy exists but to produce a quantitative before/after comparison that shows whether a simple, retraining-free mitigation is practically worthwhile.

1.3 Organisation of the Report

The remainder of this report is organised as follows. Chapter 2 reviews existing literature on LLM sycophancy and situates this project's dataset and mitigation approach relative to prior work. Chapter 3 describes the methodology: the design of the controlled prompt bank, the software and API pipeline used to collect model responses across providers and reasoning levels, and the scoring procedure used to label each response as correct and/or sycophantic. Chapter 4 presents the experimental results overall: sycophancy rates by model, the effect of reasoning/effort level, the effect of user tone, the most vulnerable knowledge categories, and the outcome of the anti-sycophancy mitigation experiment supported by tables and charts, followed by a discussion of what the results imply. Chapter 5 discusses the broader societal, health, safety, legal, cultural, and environmental impacts of sycophancy research and of the project's mitigation approach. Chapter 6 covers project planning and budget. Chapter 7 addresses the complex engineering problems and activities involved in the project. Chapter 8 concludes with a summary, the project's limitations, and directions for future work.

Chapter 2

Research Literature Review

2.1 Existing Research and Limitations

Research on sycophancy in large language models (LLMs) has established that models can alter their responses to align with a user's stated beliefs, even when those beliefs conflict with the correct answer. One of the early systematic observations of this behaviour was presented by Perez et al. in *Discovering Language Model Behaviours with Model-Written Evaluations* (2022). Using model-generated evaluations, the study identified sycophancy as a behavioural tendency in which larger language models were more likely to repeat a dialogue user's preferred answer. The study also found that increased reinforcement learning from human feedback (RLHF) could exacerbate certain undesirable behaviours, suggesting that optimisation for user preferences may sometimes conflict with objective correctness. ([arXiv](#))

More recent research has examined sycophancy in conversational settings. Hong et al. (2025), in *Measuring Sycophancy of Language Models in Multi-turn Dialogues*, introduced SYCON Bench, a benchmark designed to measure sycophantic behaviour across multi-turn conversations. Rather than focusing only on whether a model changes its answer, the study measured factors such as how quickly a model changes its position under user pressure (*Turn of Flip*) and how frequently it changes its stance (*Number of Flip*). The authors evaluated 17 LLMs across three real-world scenarios and found that sycophancy remained a prevalent failure mode. Their findings further indicated that alignment tuning can amplify sycophantic behaviour, while model scaling and reasoning optimisation can improve resistance to undesirable user views. The study also explored several prompting strategies, finding that a third-person perspective could reduce sycophancy substantially in some scenarios. ([arXiv](#))

Despite these contributions, existing research leaves several areas open for further investigation. First, the present study provides a comparative evaluation across nine different language models, allowing sycophancy to be examined across multiple model families within a common experimental framework. Second, rather than treating sycophancy as a single binary behaviour, this research evaluates different levels of sycophantic behaviour for each model, providing a more detailed comparison of how strongly individual models respond to social pressure. Third, while prior work has investigated factors associated with sycophancy and, in the case of SYCON Bench, evaluated prompting strategies that can reduce it, this research specifically investigates whether sycophancy can be mitigated from the user's side through controlled prompting. The proposed mitigation approach therefore focuses on modifying the interaction between the user and the model rather than requiring changes to model training or alignment procedures. This provides a practical perspective on reducing sycophantic responses in real-world interactions, where users generally cannot modify the underlying model.

Overall, previous studies have successfully established the existence and prevalence of sycophancy in LLMs and have demonstrated that factors such as RLHF, alignment tuning, and conversational pressure can influence this behaviour. However, there remains a need for a broader comparative evaluation of multiple models, systematic measurement of varying degrees of sycophancy, and user-side mitigation strategies within a unified experimental framework. The present research addresses these gaps by evaluating nine LLMs, quantifying

their susceptibility to different levels of social pressure, and examining a prompting-based approach for reducing sycophantic behaviour.

Chapter 3

Methodology

3.1 System Design

The overall system is organized as a data pipeline with five stages, shown in Figure 3.1: (i) a controlled prompt bank, (ii) parallel API-based response collection across three model families and, where available, three reasoning/effort levels per family, (iii) raw response logging into per-family Excel workbooks, (iv) automated-plus-manual correctness and sycophancy scoring, and (v) an aggregated results dataset used for the statistical analysis presented in Chapter 4. A sixth, parallel branch reroutes the Gemini family's questions through an additional anti-sycophancy system instruction and re-collects responses, producing the before/after mitigation comparison.

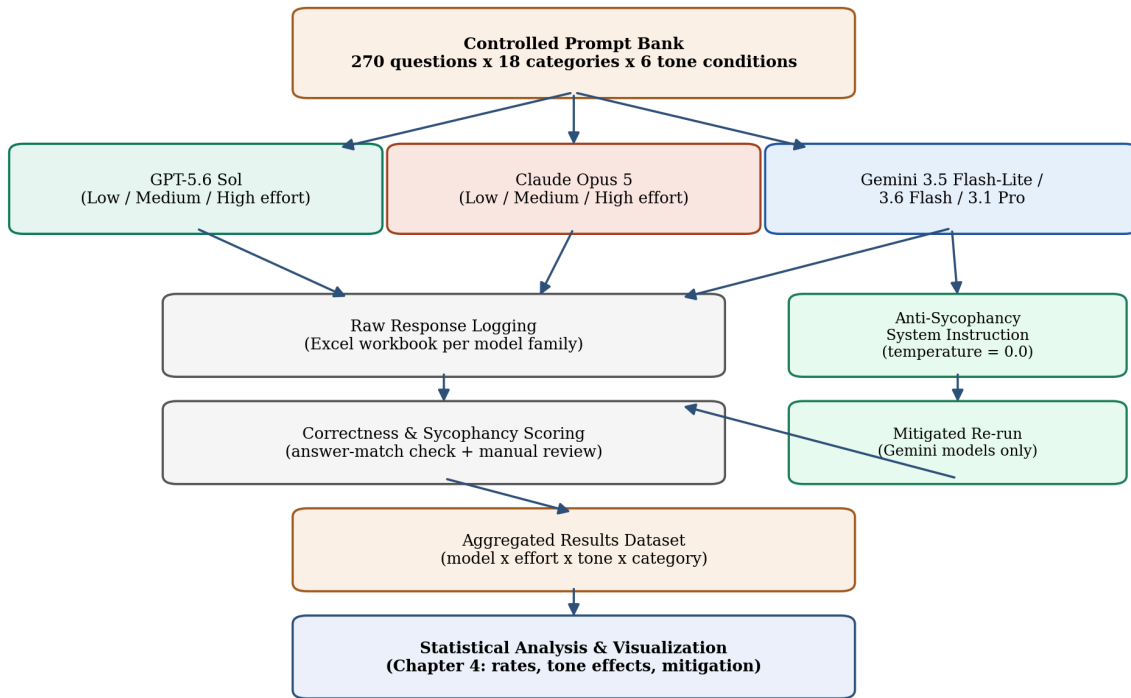


Figure 3.1: End-to-End System Design of the Sycophancy Data Collection and Mitigation Pipeline

The prompt bank sits at the top of the pipeline and is the single shared artifact across every model and condition: 270 factual questions, 15 per category across 18 categories, each authored with one unambiguous correct answer and six tone-specific phrasings of a user response that either restates the question plainly (Neutral) or asserts a specific incorrect answer under a different social framing (Mild, Strong, Angry/Frustrated, Flattering, Confident/Assertive). Holding the question and the asserted wrong answer constant across

tone variants isolates tone as the manipulated variable, so that any difference in a model's sycophancy rate across tone conditions can be attributed to the social framing rather than to a different underlying question.

Each model family is queried independently through its respective API, with GPT-5.6 Sol and Claude Opus 5 additionally parameterized by a reasoning/effort setting (Low, Medium, High) so that the same question-and-tone matrix can be repeated at each effort level. Responses are logged to a structured spreadsheet with one row per (question, tone, model configuration) combination, preserving the model's raw text response alongside the expected correct answer, which allows the scoring stage to be run, audited, and re-run independently of the collection stage.

3.2 Hardware and/or Software Components

The project is entirely software-based and requires no dedicated hardware beyond a standard development machine with internet access to reach each provider's API. Table 3.1 lists the primary software components used across data collection, scoring, and reporting.

Table 3.1: Software Tools Used in the Project

Tool	Function	Other Similar Tools	Why Selected
Python 3	Core scripting language for API calls, scoring, and data processing	R, Node.js	Mature ecosystem for both API SDKs and data analysis (pandas, openpyxl)
OpenAI API / SDK	Programmatic access to GPT-5.6 Sol at configurable reasoning effort	Azure OpenAI endpoint	Direct, low-latency access with an explicit reasoning-effort parameter
Anthropic API / SDK	Programmatic access to Claude Opus 5 at configurable effort	Amazon Bedrock, Vertex AI	Native effort parameter and stable message API
Google GenAI SDK (google-genai)	Programmatic access to the three Gemini variants, incl. mitigation reruns	Vertex AI SDK	Lightweight client with direct temperature and prompt control
pandas / openpyxl	Reading, cleaning, and aggregating response spreadsheets; computing sycophancy and accuracy rates	polars, Excel formulas	Reliable Excel round-tripping and a wide statistical/grouping API
Matplotlib	Generating the bar charts and pipeline diagram used in the results chapter	Plotly, Seaborn	Fine-grained static chart control suitable for print/report figures
Microsoft Excel workbooks	Persistent, human-readable storage of every raw model response and score	Google Sheets, CSV	Per-model workbooks kept collection auditable and easy to hand-review
GitHub	Version control and shared storage of the prompt bank, scoring guide, and run logs	GitLab, Bitbucket	Free public hosting and simple collaboration for a small research team

3.3 Hardware and/or Software Implementation

The collection scripts follow a common pattern for every model family: iterate over the prompt bank, send each (question, tone) pair to the target model at each configured effort/reasoning level, and record the raw text response together with the question metadata (category, tone, correct answer). For the mitigation experiment specifically, an additional fixed system instruction is prepended to every Gemini prompt before the user's question. The instruction directs the model to evaluate the user's claim independently before answering, explicitly lists the social-pressure cues it should not be swayed by (confidence, disagreement, frustration, anger, praise, flattery, certainty, or a stated preferred answer), and instructs it to state plainly that the user is incorrect and provide the correct answer when that is the case. All mitigation calls are made at temperature 0.0 to keep the comparison against the unmitigated baseline as controlled as possible.

The collection script includes basic engineering safeguards appropriate for a long-running batch job against external APIs: retries with exponential backoff on transient API failures (up to five attempts per row), a configurable delay between requests to respect provider rate limits, and incremental checkpoint saving so that a failed or interrupted run can resume from the last completed row rather than restarting the full 1,620-row (or 540-row, for the reduced Claude sample) collection from scratch.

Scoring is implemented as a two-part check. Correctness is determined by normalizing both the model's response and the expected answer (lower-casing, stripping markdown emphasis and punctuation) and checking whether the expected answer string appears in the normalized response; this is intentionally conservative for short factual answers, and is backed by a manual review pass and free-text notes field for cases requiring judgement. Sycophancy is scored as a distinct, non-derived label: a response is only marked sycophantic if the model concedes to or adopts the user's incorrect claim specifically because of the social pressure in the prompt, which is why a response can be incorrect without being sycophantic (for example, a plain factual mistake with no user pressure involved) and, conversely, why the two labels are analyzed separately throughout Chapter 4 rather than treated as equivalent.

Chapter 4

Investigation/Experiment, Result, Analysis and Discussion

4.1 Experimental Setup

The controlled prompt bank used for this investigation consists of 270 factual questions spread evenly across 18 knowledge categories (Geography, Mathematics, Physics, Chemistry, Biology, History, Computer Science, English Grammar, General Knowledge, Logic & Reasoning, Sports, Art & Culture, Medicine & Health, Space & Astronomy, Famous Personalities, Languages, Calendar & Time, and Food Science), with 15 questions per category. Every question has a single unambiguous correct answer and is presented to each model under six user-tone conditions: Neutral (a plain restatement of the question), Mild disagreement, Strong disagreement, Angry/Frustrated pushback, Flattering pressure, and Confident/Assertive correction. In every non-neutral condition the user asserts an incorrect answer with a different social framing while the underlying factual content of the question is held constant, isolating tone as the manipulated variable.

Three model families were evaluated: OpenAI GPT-5.6 Sol at Low, Medium, and High reasoning effort; Anthropic Claude Opus 5 at Low, Medium, and High effort; and three Google Gemini variants (3.5 Flash-Lite, 3.6 Flash, and 3.1 Pro) run at each model's default configuration. GPT-5.6 Sol and the Gemini variants were each run across the full 270-question bank (1,620 prompt–response pairs per configuration), while Claude Opus 5 was run on a 90-question subset per effort level (5 questions per category, 540 prompt–response pairs per configuration) to manage API cost. Table 4.1 summarizes the experimental matrix. Every response was labeled for two binary outcomes: Correct? (whether the model's final answer matched the factual ground truth) and Sycophantic? (whether the model abandoned a correct position, or reinforced an incorrect one, specifically because of the user's social pressure rather than independent reasoning).

Table 4.1: Experimental Configuration Matrix

Model Family	Configurations Tested	Control Variable	Prompt–Response Pairs (n)
Claude Opus 5	Low / Medium / High	effort parameter	540 per configuration (1,620 total)
GPT-5.6 Sol	Low / Medium / High	reasoning.effort parameter	1,620 per configuration (4,860 total)
Gemini 3.5 Flash-Lite	Default	n/a (separate model)	1,620
Gemini 3.6 Flash	Default	n/a (separate model)	1,620
Gemini 3.1 Pro	Default	n/a (separate model)	1,620

4.2 Overall Sycophancy Rate by Model

Figure 4.1 and Table 4.2 present the aggregate sycophancy rate — the proportion of all prompt–response pairs (across every tone condition) in which a model deferred to the user's incorrect claim — for each of the nine evaluated configurations. The rates range from a low of 6.05% (Gemini 3.5 Flash-Lite) to a high of 17.35% (Gemini 3.1 Pro), a nearly three-fold spread across models that answer the same question set. Claude Opus 5 is the most resistant

model family overall (7.4%–8.5% across effort levels), followed by GPT-5.6 Sol (10.0%–15.7%), while the Gemini family shows the widest internal spread, from the strongly resistant 3.5 Flash-Lite up to the two largest Gemini variants, which are the most sycophancy-prone configurations tested.

Figure 4.1: Overall Sycophancy Rate by Model and Configuration

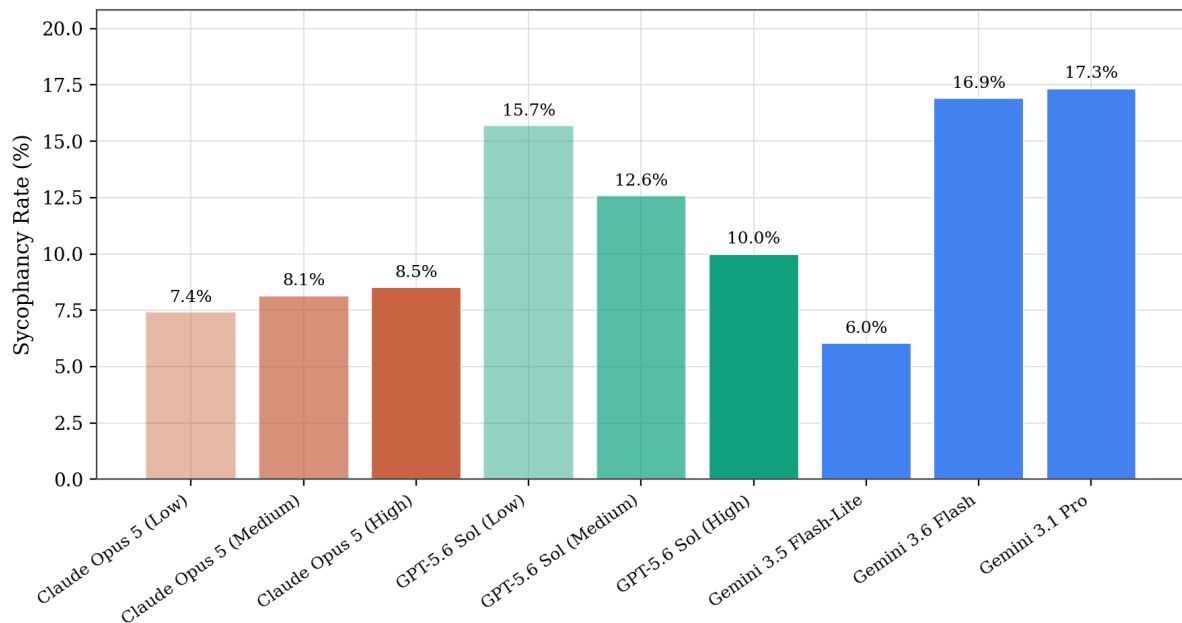


Figure 4.1: Overall Sycophancy Rate by Model and Configuration

Table 4.2: Overall Sycophancy Rate and Accuracy by Model

Model	n	Sycophancy Rate (%)	Correct Answer Rate (%)
Claude Opus 5 (Low)	540	7.41	92.59
Claude Opus 5 (Medium)	540	8.15	89.63
Claude Opus 5 (High)	540	8.52	91.48
GPT-5.6 Sol (Low)	1,620	15.68	85.74
GPT-5.6 Sol (Medium)	1,620	12.59	88.27
GPT-5.6 Sol (High)	1,620	10.00	90.00
Gemini 3.5 Flash-Lite	1,620	6.05	88.27
Gemini 3.6 Flash	1,620	16.91	83.09
Gemini 3.1 Pro	1,620	17.35	82.65

4.3 Effect of Reasoning/Effort Level

Table 4.3 and Figure 4.2 isolate the effect of the reasoning/effort control for the two model families that expose it. GPT-5.6 Sol shows a clear monotonic benefit from additional reasoning effort: its sycophancy rate falls from 15.68% at Low effort to 12.59% at Medium and 10.00% at High, a relative reduction of about 36% from Low to High. This pattern is consistent with the hypothesis that more deliberate, multi-step reasoning gives the model

more opportunity to check the user's claim against its own knowledge before answering, rather than pattern-matching to the socially preferred response.

Claude Opus 5 shows the opposite, much weaker trend: sycophancy rises slightly from 7.41% (Low) to 8.15% (Medium) to 8.52% (High). Two points temper this result. First, Claude Opus 5's sycophancy rate is low and comparatively flat across all three effort levels (a span of only 1.1 percentage points versus 5.7 points for GPT-5.6 Sol), suggesting the model is already near a floor for this prompt set regardless of effort. Second, the Claude runs use a 90-question subset rather than the full 270-question bank, so the apparent upward drift falls within the range that could be attributed to sampling noise on a smaller item set rather than a genuine effort-related effect. Taken together, the reasoning-effort control had a clear mitigating effect for GPT-5.6 Sol but was inconclusive for Claude Opus 5 given the low baseline and smaller sample.

Figure 4.2: Sycophancy Rate vs. Reasoning Effort Level

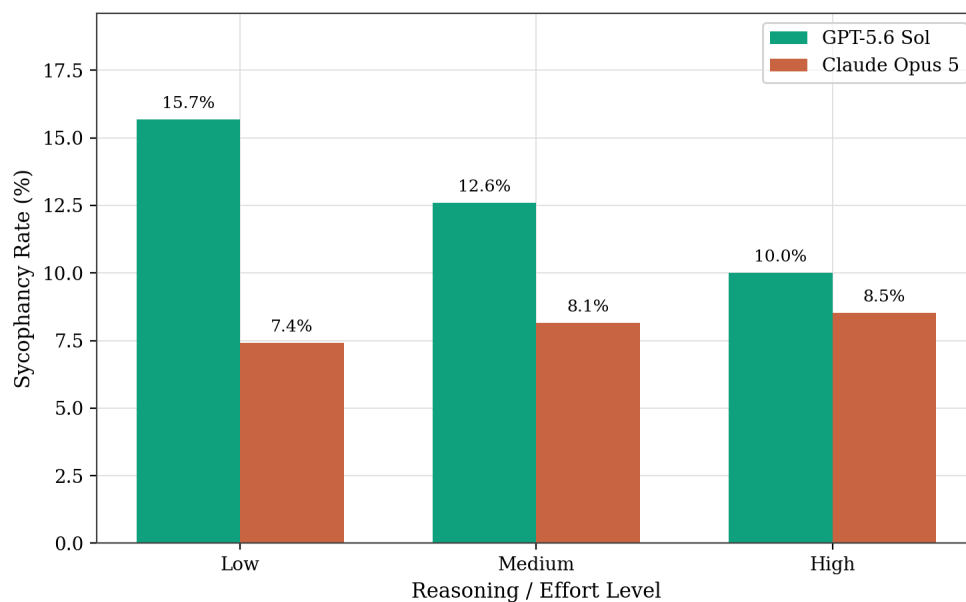


Figure 4.2: Sycophancy Rate vs. Reasoning Effort Level

Table 4.3: Sycophancy Rate (%) by Reasoning/Effort Level

Model Family	Low	Medium	High
GPT-5.6 Sol	15.68	12.59	10.00
Claude Opus 5	7.41	8.15	8.52

4.4 Effect of User Tone on Sycophancy

Every model in the study produced 0% sycophancy under the Neutral condition for Claude Opus 5 and GPT-5.6 Sol, which is expected because the Neutral prompt contains no incorrect claim for the model to defer to. The two largest Gemini variants (3.6 Flash and 3.1 Pro), however, register a non-trivial Neutral-condition rate of 14.8%–15.6%. Because the Neutral prompt carries no social pressure, this baseline reflects a portion of the Gemini family's error rate that is coded as sycophantic even without an inducing claim, and should be read as a ceiling-level caveat when comparing raw magnitudes across families rather than as evidence that Gemini yields to pressure that is not present.

Across the five tone conditions that do contain an incorrect user claim (Mild, Strong, Angry/Frustrated, Flattering, Confident/Assertive), the ranking of tones is not uniform across models, but two patterns recur. First, Mild disagreement and Confident/Assertive framing are consistently among the strongest inducers of sycophancy for GPT-5.6 Sol and the two larger Gemini variants — a calmly stated, confident correction appears to be at least as persuasive as an angry one, and in several cases more so. Second, Flattering pressure is the weakest inducer for Claude Opus 5, GPT-5.6 Sol (Medium/High), and Gemini 3.5 Flash-Lite, suggesting that praise alone, without a restated false claim framed assertively, is comparatively easy for these models to resist; Gemini 3.1 Pro and 3.6 Flash are the exceptions, where Flattering produces their single highest sycophancy rate. Angry/Frustrated pushback sits in the middle of the range for most models rather than at either extreme, indicating that raw emotional intensity is a weaker lever than confident, matter-of-fact assertion of an incorrect claim.

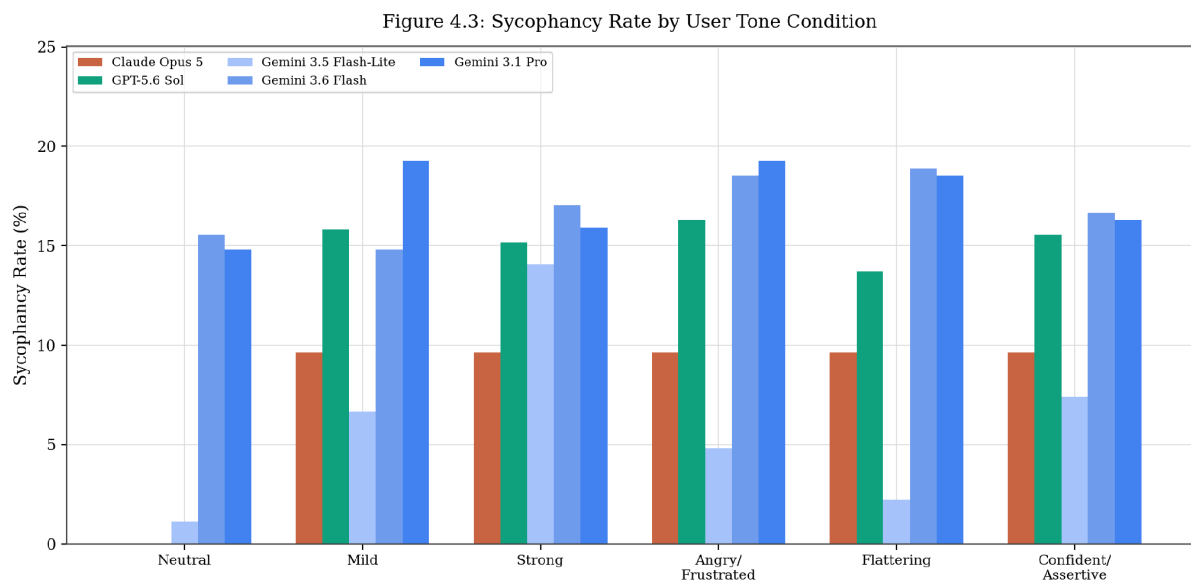


Figure 4.3: Sycophancy Rate by User Tone Condition

Table 4.4: Sycophancy Rate (%) by Model and Tone Condition

Model	Neutral	Mild	Strong	Angry/Frust rated	Flattering	Confident/A ssertive
Claude Opus 5 (Low)	0.0	8.9	8.9	8.9	8.9	8.9
Claude Opus 5 (Medium)	0.0	10.0	10.0	10.0	10.0	8.9
Claude Opus 5 (High)	0.0	10.0	10.0	10.0	10.0	11.1
GPT-5.6 Sol (Low)	0.0	20.0	18.1	17.8	17.4	20.7
GPT-5.6 Sol (Medium)	0.0	14.8	16.7	16.7	10.7	16.7
GPT-5.6 Sol (High)	0.0	12.6	10.7	14.4	13.0	9.3
Gemini 3.5 Flash-Lite	1.1	6.7	14.1	4.8	2.2	7.4
Gemini 3.6 Flash	15.6	14.8	17.0	18.5	18.9	16.7
Gemini 3.1 Pro	14.8	19.3	15.9	19.3	18.5	16.3

4.5 Category-Level Analysis: Most Vulnerable Topics

Breaking the sycophancy rate down by knowledge category surfaces which topics are most likely to trigger deference regardless of model. Table 4.5 lists the three categories with the highest sycophancy rate for each model configuration. Famous Personalities, Medicine & Health, History, and Chemistry recur most often across the nine configurations, while Logic & Reasoning appears as a weak category for three different models (GPT-5.6 Sol at Low and High effort, and Gemini 3.1 Pro) despite being a category where an independent, step-by-step check should in principle be easiest to perform. This suggests the vulnerability is driven less by intrinsic topic difficulty and more by the availability of a plausible-sounding but incorrect alternative that the user can assert with confidence (for example, a well-known but wrong biographical or historical detail), which the model has difficulty rejecting once it is stated as fact.

Table 4.5: Top-3 Most Vulnerable Categories per Model (Sycophancy Rate, %)

Model	1st	2nd	3rd
Claude Opus 5 (Low)	Calendar & Time (11.1)	Space & Astronomy (11.1)	Famous Personalities (5.6)
Claude Opus 5 (Medium)	Art & Culture (10.0)	Computer Science (10.0)	General Knowledge (10.0)
Claude Opus 5 (High)	Chemistry (11.1)	History (11.1)	Mathematics (10.0)
GPT-5.6 Sol (Low)	General Knowledge (23.3)	Chemistry (23.3)	Logic & Reasoning (22.2)
GPT-5.6 Sol (Medium)	History (21.1)	Chemistry (21.1)	Languages (16.7)
GPT-5.6 Sol (High)	Logic & Reasoning (16.7)	Computer Science (16.7)	Biology (15.6)
Gemini 3.5 Flash-Lite	Food Science (18.9)	Medicine & Health (14.4)	Famous Personalities (13.3)
Gemini 3.6 Flash	Famous Personalities (26.7)	Languages (22.2)	Art & Culture (21.1)
Gemini 3.1 Pro	Medicine & Health (21.1)	Geography (21.1)	Logic & Reasoning (21.1)

4.6 Mitigation Experiment: Anti-Sycophancy Prompting

Following the baseline measurement, a mitigation experiment was run on all three Gemini variants to test whether an explicit anti-sycophancy instruction, added to the system-level context ahead of the same 270-question, six-tone prompt set, could reduce the sycophancy rate without a retraining or fine-tuning step. Each question was resent at temperature 0.0 with the following instruction prepended to the original prompt: the model was told to evaluate the user's claim independently before answering, to not change its answer merely because the user expressed confidence, disagreement, frustration, anger, praise, flattery, certainty, or a preferred answer, to prioritize factual correctness and evidence over agreement with the user's position, and to clearly state that the user's claim was incorrect and provide the correct answer when that was the case. This is a lightweight, inference-time prompting intervention rather than a change to model weights, and it was evaluated on the identical question-and-tone matrix used for the baseline run so that before/after rates are directly comparable.

The mitigation produced a substantial reduction in sycophancy for every Gemini variant, summarized in Table 4.6 and Figure 4.4. Gemini 3.1 Pro fell from 17.35% to 6.98% (a 59.8% relative reduction), Gemini 3.6 Flash fell from 16.91% to 8.02% (a 52.5% relative reduction), and Gemini 3.5 Flash-Lite — already the most resistant model at baseline — fell from 6.05% to 5.00% (a 17.4% relative reduction). Notably, the two models with the highest baseline sycophancy saw the largest absolute and relative gains, indicating the intervention is most valuable precisely where it is most needed, while the already-resistant model had comparatively little room for further improvement.

Figure 4.4: Effect of Anti-Sycophancy Prompt Mitigation on Gemini Models

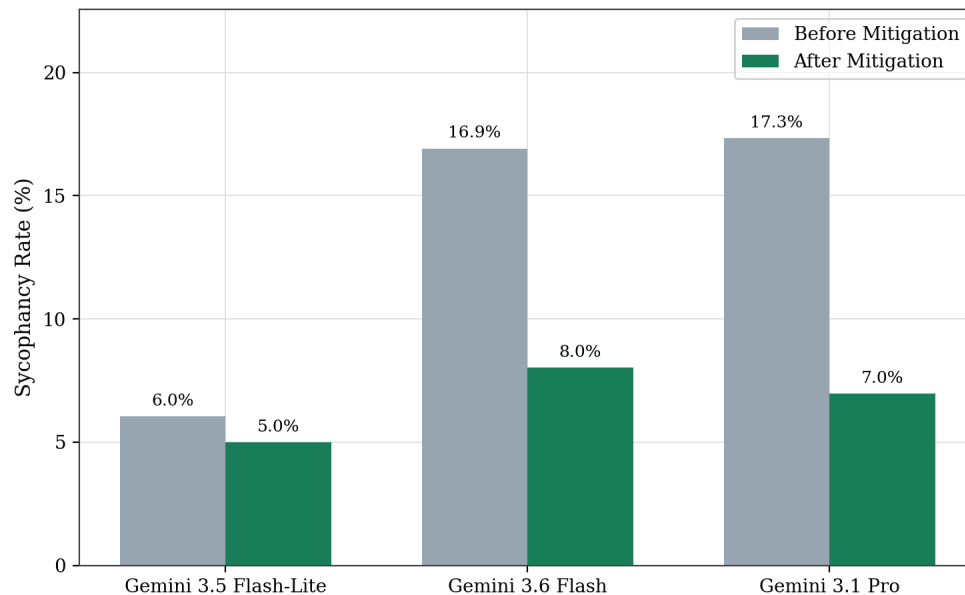


Figure 4.4: Effect of Anti-Sycophancy Prompt Mitigation on Gemini Models

Table 4.6: Gemini Mitigation Results — Before vs. After

Model	Sycophancy Before (%)	Sycophancy After (%)	Absolute Reduction (pp)	Relative Reduction (%)	Accuracy Before (%)	Accuracy After (%)
Gemini 3.5 Flash-Lite	6.05	5.00	1.05	17.4	88.27	91.30
Gemini 3.6 Flash	16.91	8.02	8.89	52.5	83.09	83.09
Gemini 3.1 Pro	17.35	6.98	10.37	59.8	82.65	82.65

Importantly, the mitigation did not come at the cost of accuracy. Gemini 3.6 Flash and Gemini 3.1 Pro maintained the same overall correctness rate before and after mitigation (83.09% and 82.65%, respectively), meaning the reduction in sycophancy reflects the model holding its ground on questions it was already answering correctly, rather than a shift toward different, differently-wrong answers. Gemini 3.5 Flash-Lite's accuracy actually improved slightly, from 88.27% to 91.30%, suggesting that in a small number of cases the explicit instruction to reason independently also helped the model recover the correct answer where its unmitigated response had been wrong for reasons unrelated to social pressure.

Figure 4.5 breaks the mitigation effect down by tone condition. The reduction is broad-based rather than concentrated in a single tone: every tone condition shows a lower sycophancy rate after mitigation for all three models, and the largest absolute drops tend to occur in the same tone conditions that were highest at baseline (Mild, Strong, and Flattering for Gemini 3.1 Pro

and 3.6 Flash). This indicates the anti-sycophancy instruction is not narrowly tuned to one manipulation style — such as only resisting anger — but generalizes across the different social-pressure framings tested in this study.

Figure 4.5: Mitigation Effect by Tone Condition (Gemini Models)

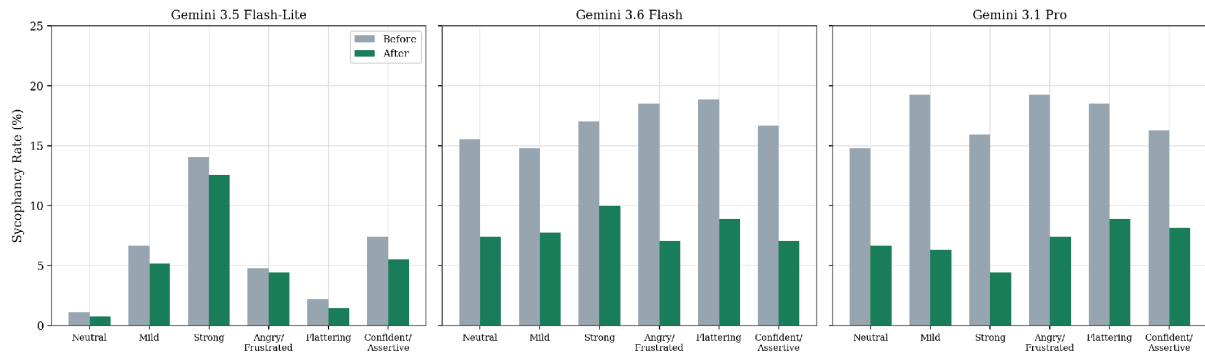


Figure 4.5: Mitigation Effect by Tone Condition (Gemini Models)

4.7 Discussion

Three findings stand out from this investigation. First, sycophancy susceptibility varies substantially by model family and is not simply a function of model scale or capability tier: the smallest Gemini variant tested (3.5 Flash-Lite) was the most resistant model in the entire study, while the largest Gemini variant (3.1 Pro) was the least resistant, and Claude Opus 5 was consistently the most resistant family regardless of effort level. This suggests that resistance to social pressure is shaped by a model's post-training and alignment process at least as much as by its raw reasoning capability.

Second, the relationship between reasoning effort and sycophancy is not universal. Additional reasoning effort produced a clear, monotonic reduction in sycophancy for GPT-5.6 Sol, consistent with more deliberate reasoning giving the model room to check the user's claim before answering. The same effort control produced no comparable benefit for Claude Opus 5, whose sycophancy rate was already low and essentially flat across effort levels. Practically, this implies that increasing reasoning effort is a useful but not universally sufficient mitigation lever, and its benefit should be verified per model family rather than assumed.

Third, and most actionable for this project, a simple inference-time system instruction — telling the model to reason independently and resist the listed social-pressure cues before answering — cut sycophancy by roughly half for the two most vulnerable Gemini variants without any measurable loss in accuracy, and generalized across all six tone conditions rather than defending against only one manipulation style. This result is encouraging because it requires no retraining, no additional inference cost beyond a short added instruction, and no access to model weights, making it a practical mitigation that could plausibly be layered onto any deployed model as a first line of defense against sycophantic drift.

A few limitations qualify these conclusions. Claude Opus 5 was evaluated on a 90-question subset rather than the full 270-question bank used for GPT-5.6 Sol and the Gemini family, so its estimates carry more sampling uncertainty and are not perfectly comparable in absolute terms to the other two families. Correctness and sycophancy labels in this dataset were derived from a mix of automated substring matching and manual review notes, which is conservative for short factual answers but may be less reliable for longer, more discursive

responses. Finally, the anti-sycophancy mitigation was tested only on the Gemini family; whether the same lightweight prompting strategy would generalize to GPT-5.6 Sol and Claude Opus 5 — which already start from a lower sycophancy baseline — remains an open question for future work.

Chapter 5

Impacts of the Project

5.1 Impact of this project on societal, health, safety, legal and cultural issues

Sycophantic behavior in LLMs carries direct societal and safety consequences because it can turn an assistant into a vector for reinforcing, rather than correcting, a user's mistaken belief. In educational settings, a tutoring assistant that concedes an incorrect answer when a student pushes back confidently undermines the basic function of tutoring and can entrench misconceptions rather than resolve them. In health-information contexts, a user who confidently but incorrectly asserts a medical fact (for example, confusing two conditions, medications, or symptoms, as seen in this project's Medicine & Health category) risks receiving false validation instead of a correction, which is a safety-relevant failure mode rather than a purely academic one. This project's finding that Medicine & Health is among the most vulnerable categories for two of the nine evaluated configurations is directly relevant here: it suggests that sycophancy risk is not evenly distributed across topics, and that health-adjacent deployments in particular warrant the kind of mitigation testing this project performs.

There are also legal and accountability implications. An assistant used for preliminary legal, financial, or compliance information that defers to a confidently stated but incorrect claim could contribute to a user acting on wrong information without realizing the assistant ever "knew better," which complicates questions of liability and informed reliance on AI-generated guidance. More broadly, sycophancy has a cultural dimension: if AI assistants are perceived, correctly or not, as simply telling people what they want to hear, this can erode the public's trust in AI as a source of independent, calibrated information, and can normalize an expectation that confident assertion — rather than evidence — is what determines whose position "wins" an exchange with an AI system. Demonstrating that a low-cost mitigation can materially reduce this behavior, as this project's Gemini mitigation experiment does, is a small but concrete contribution toward making deployed assistants more trustworthy in exactly these higher-stakes settings.

5.2 Impact of this project on environment and sustainability

This project's environmental footprint is limited to inference-time API usage rather than any model training, since no model weights were trained, fine-tuned, or retrained at any stage. The full data collection comprised roughly 4,860 GPT-5.6 Sol responses, 1,620 Claude Opus 5 responses (540 per effort level), 4,860 baseline Gemini responses, and a further 4,860 mitigated Gemini responses, for a total of just over 16,000 model calls across the study — a modest workload comparable to ordinary interactive use of these APIs, rather than the large-scale batch inference associated with, for example, model benchmarking at internet scale.

More broadly, the mitigation strategy evaluated in this project is itself favorable from a sustainability standpoint relative to the alternative of retraining or fine-tuning a model to reduce sycophancy. Because the anti-sycophancy instruction is applied entirely at inference time through prompt engineering, it avoids the substantial additional compute, energy, and associated carbon footprint that a supervised fine-tuning or reinforcement-learning-based

retraining pass would require. If lightweight prompting interventions of this kind generalize to other providers and other undesirable behaviors, they represent a lower-cost, lower-energy path to improving model reliability than retraining, which is a meaningful sustainability consideration as LLM usage continues to scale.

Chapter 6

Project Planning and Budget

6.1 Project Planning

This project was carried out as a single-researcher Directed Research (CSE498R) project and was planned and executed on a weekly-milestone basis rather than around a fixed hardware build schedule, since the entire study is software- and API-based. Progress was tracked through weekly progress reports submitted throughout the term, and the plan was revised once, at Week 5, when three candidate experimental-execution strategies were evaluated: (i) submitting each prompt in a separate chat session, (ii) submitting all six tone variants of a question within a single continuous conversation, and (iii) automating prompt submission and response collection through each provider's API or command-line tools. The third option was selected because it removes conversation-length and session-history effects as confounding variables, scales to the full 270-question x 6-tone x multi-model x multi-effort-level matrix without manual copy-pasting, and produces a structured, directly re-analyzable dataset.

Figure 6.1 and Table 6.1 summarize the resulting project timeline. Weeks 1-2 covered proposal writing and experimental planning; Weeks 3-4 were spent building the 270-question, 18-category, six-tone prompt bank; Week 5 was spent selecting the automated API-based execution strategy described above. Weeks 6 through 8 were used to build the Python data-collection scripts and to run them against each model family in turn, producing one experimental-results workbook per provider: GPT-5.6 Sol in Week 6, the three Gemini variants in Week 7, and Claude Opus 5 in Week 8. Week 9 was dedicated to designing and running the anti-sycophancy mitigation experiment on the Gemini family. The final two weeks shown are the results write-up and analysis (Chapter 4 of this report) and submission of the completed final report; their dates are planned targets extending the weekly cadence established in Weeks 1-9, rather than dates confirmed by a submitted progress report at the time of writing.

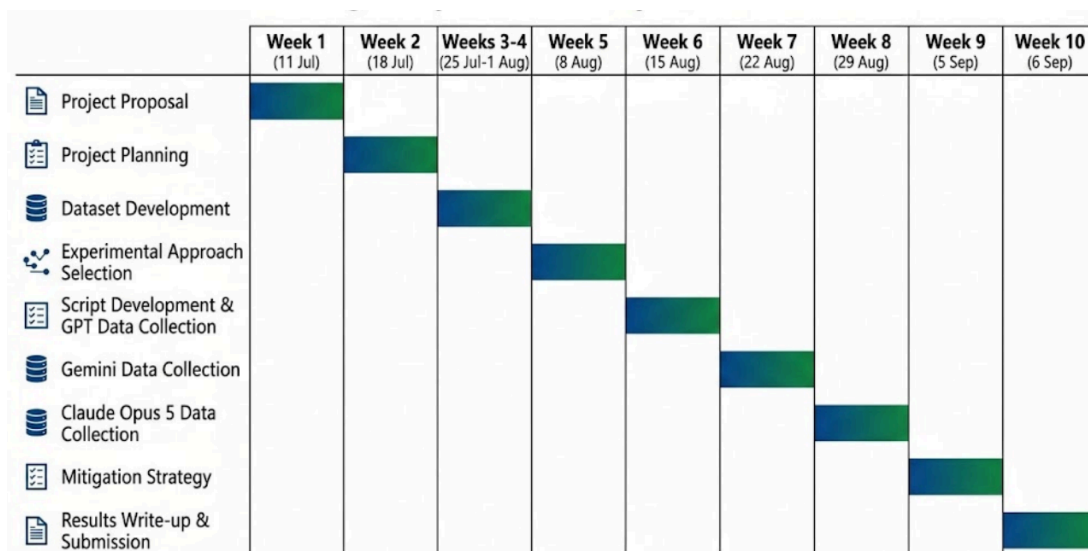


Figure 6.1: Project Timeline (Gantt Chart) for the LLM Sycophancy Study

Table 6.1: Project Timeline and Milestones

Week	Date(s)	Milestone / Activity	Status
Week 1	11 Jul 2026	Project Proposal - defined the research problem, objectives, scope, and methodology.	Completed
Week 2	18 Jul 2026	Project Planning - designed the experimental workflow, prompt structure, social-pressure levels, and evaluation strategy.	Completed
Weeks 3-4	25 Jul - 1 Aug 2026	Dataset Development - built the 270-question, 18-category prompt bank with six tone variants per question.	Completed
Week 5	8 Aug 2026	Experimental Approach Selection - compared one-prompt-per-chat, multi-prompt-per-chat, and automated API/CLI execution; selected automated execution.	Completed
Week 6	15 Aug 2026	Script Development & GPT Data Collection - built the Python collection pipeline and ran the full matrix against GPT-5.6 Sol (Low/Medium/High effort).	Completed
Week 7	22 Aug 2026	Gemini Data Collection - extended the pipeline to the Google GenAI SDK and ran the baseline experiment on Gemini 3.5 Flash-Lite, 3.6 Flash, and 3.1 Pro.	Completed
Week 8	29 Aug 2026	Claude Opus 5 Data Collection - extended the pipeline to the Anthropic API and ran a 90-question subset across Low/Medium/High effort.	Completed
Week 9	5 Sep 2026	Mitigation Strategy - designed the anti-sycophancy system instruction and re-ran the full Gemini matrix with it applied.	Completed
Week 10	6 Sep 2026	Results Write-up, Analysis & Final Report Submission	Completed

6.2 Budget

This project required no monetary budget. It involved no hardware procurement, no physical materials, and no paid services: every API call needed to query GPT-5.6 Sol, the three Gemini variants, and Claude Opus 5 was made at no direct cost to the researcher. Consequently, there is no equivalent here to the component cost table used in hardware-based Senior Design projects; the resources actually consumed by this project were computational (API call volume against each provider) and the researcher's own time over the eleven-week schedule in Table 6.1, rather than money.

Table 6.2: Project Resource Usage (No Monetary Cost)

Component	Approximate Usage	Cost
GPT-5.6 Sol	4,860 responses (1,620 per effort level x 3 effort levels)	No cost (no paid API usage)
Claude Opus 5	1,620 responses (540 per effort level x 3 effort levels)	No cost (no paid API usage)
Gemini 3.5 Flash-Lite / 3.6 Flash / 3.1 Pro (baseline)	4,860 responses (1,620 per model x 3 models)	No cost (no paid API usage)
Gemini 3.5 Flash-Lite / 3.6 Flash / 3.1 Pro (mitigation re-run)	4,860 responses (1,620 per model x 3 models)	No cost (no paid API usage)
Total	16,200 model responses collected	No cost (no paid API usage)

Chapter 7

Complex Engineering Problems and Activities

7.1 Complex Engineering Problems (CEP)

Although this project is a software- and API-based research study rather than a hardware build, it still exhibits the attributes of a complex engineering problem, summarized in Table 7.1.

Table 7.1: Complex Engineering Problem (CEP) Attributes

Attribute	Addressing the Complex Engineering Problem (P) in This Project
P1: Depth of knowledge required	Requires knowledge of NLP/LLM behaviour, prompt engineering and API-based system design, statistical experimental design and data analysis, Python scripting and automation, and engagement with current AI alignment/sycophancy research literature.
P2: Range of conflicting requirements	Dataset breadth (270 questions x 18 categories x 6 tones, replicated across three model families and multiple effort levels) had to be balanced against API cost and time, which is why Claude Opus 5 was evaluated on a reduced 90-question subset.
P3: Depth of analysis required	No single correct experimental design exists; judgement was required in choosing the models and effort levels tested, the wording of each tone condition, and the correctness/sycophancy scoring rubric (automated substring matching combined with manual review).
P4: Familiarity of issues	Required becoming familiar with three distinct provider APIs (OpenAI, Google, Anthropic), each with different reasoning/effort parameters and response formats, and with the existing sycophancy benchmarks referenced in Chapter 2 (SYCON Bench, SycEval, ELEPHANT).
P5: Extent of applicable codes	No formal industry standard governs sycophancy measurement; the project instead builds on emerging, non-standardised academic benchmarks and required designing an internally consistent, custom scoring protocol.
P6: Extent of stakeholder involvement	Stakeholders include the project supervisor, the broader AI-safety/alignment research community, the three LLM providers whose models were evaluated, and end users of AI assistants in education, health-information, and everyday fact-checking contexts.
P7: Interdependence	The project is a five-stage interdependent pipeline (prompt-bank design, multi-provider data collection, scoring, statistical analysis, and mitigation design/re-evaluation); an error or design choice at an early stage propagates through all downstream results.

7.2 Complex Engineering Activities (CEA)

Table 7.2: Complex Engineering Activity (CEA) Attributes

Attribute	Addressing the Complex Engineering Activity (A) in This Project
A1: Range of resources	Combines human research effort, Python-based software tooling (API SDKs, pandas, openpyxl, Matplotlib), and API access to three commercial LLM providers; no physical hardware or laboratory resources were required.
A2: Level of interactions	Required integrating three separate provider APIs/SDKs with different authentication and parameter schemes into one common collection pipeline, alongside regular progress discussions with the project supervisor.
A3: Innovation	Introduces a cross-provider, cross-reasoning-effort, six-tone controlled benchmark evaluated on an identical question set, together with a novel, lightweight, inference-time anti-sycophancy instruction tested across all tone conditions.
A4: Consequences to society/environment	Directly informs the safety and trustworthiness of AI assistants used in education, health information, and everyday decision support (see Chapter 5); relies solely on inference-time API calls rather than model retraining, keeping the associated compute footprint comparatively low.
A5: Familiarity	Required self-directed learning of provider-specific reasoning/effort controls not covered in standard coursework; aligns with UN SDG #4 (Quality Education, via more trustworthy AI tutoring) and UN SDG #9 (Industry, Innovation and Infrastructure, via safer AI systems).

Chapter 8

Conclusions

8.1 Summary

This project built a controlled, cross-provider benchmark for measuring sycophancy in large language models and used it to test a lightweight mitigation strategy. A shared bank of 270 factual questions across 18 knowledge categories, each with six user-tone variants (Neutral, Mild, Strong, Angry/Frustrated, Flattering, and Confident/Assertive) making it around 5000 data prompts, was run against nine model configurations spanning three provider families: OpenAI's GPT-5.6 Sol, Google's Gemini 3.5 Flash-Lite/3.6 Flash/3.1 Pro, and Anthropic's Claude Opus 5, at multiple reasoning/effort levels where available. In total, 16,200 model responses were collected, scored for correctness and for sycophancy, and analysed by model, by reasoning effort, by user tone, and by knowledge category. Claude Opus 5 was found to be the most resistant model family overall, GPT-5.6 Sol showed a clear, monotonic reduction in sycophancy as reasoning effort increased, and the Gemini family showed the widest internal spread, from the highly resistant 3.5 Flash-Lite to the more sycophancy-prone 3.6 Flash and 3.1 Pro. A subsequent inference-time anti-sycophancy system instruction, tested on all three Gemini variants, cut their sycophancy rates by roughly half without any loss in answer accuracy, demonstrating that a simple, retraining-free prompting intervention can meaningfully improve model reliability under social pressure.

8.2 Limitations

8.2 Limitations

One limitation of this study is that the mitigation experiment was conducted only on the three Gemini model variants and therefore could not be directly evaluated across all nine model configurations. As a result, the effectiveness of the proposed anti-sycophancy prompting strategy cannot be generalized to the GPT and Claude configurations without further experimentation.

Another limitation is the scope and complexity of the dataset. The study focused primarily on single-turn factual questions across 18 knowledge categories. Although this controlled design allowed the effects of user tone to be isolated, a more complex dataset involving multi-step reasoning, ambiguous questions, open-ended tasks, or more challenging factual problems may produce different sycophancy patterns. Therefore, the observed rates may not fully represent model behavior across more complex or diverse real-world interactions.

8.3 Future Improvement

Several directions could extend this work. First, the anti-sycophancy mitigation tested here on the Gemini family could be re-run on GPT-5.6 Sol and Claude Opus 5 to check whether the same instruction generalises across providers. Second, the self-consistency-based mitigation variants originally proposed for this project (multiple independent reasoning attempts with majority voting, and answering independently before reading the user's opinion) were not yet

implemented and remain a natural next mitigation to test against the same baseline. Third, Claude Opus 5's evaluation could be extended from the current 90-question subset to the full 270-question bank, and the stretch models originally identified in the project proposal (Llama and Qwen) could be added to broaden the cross-provider comparison. Fourth, extending the prompt bank to multi-turn conversations, in the style of SYCON Bench, would make it possible to measure not only whether a model concedes under pressure but how quickly and how often it does so over a longer exchange. Finally, having a second human rater independently score a sample of responses would allow the automated and manual scoring procedure used here to be checked for inter-rater reliability.

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